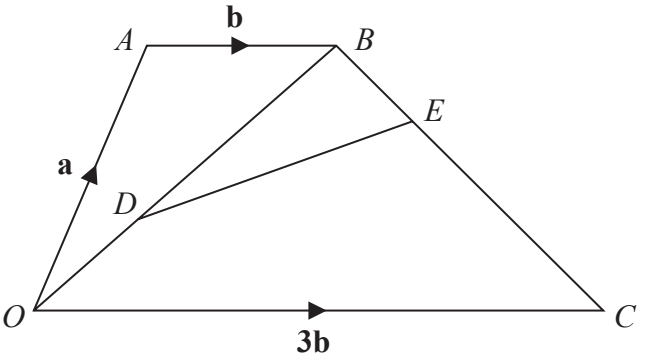
18 $OABC$ is a trapezium.



$\overrightarrow{OA} = \mathbf{a}$
$\overrightarrow{AB} = \mathbf{b}$
$\overrightarrow{OC} = \mathbf{3b}$

D is the point on OB such that $OD : DB = 2 : 3$
E is the point on BC such that $BE : EC = 1 : 4$

Work out the vector $\overrightarrow{DE}$ in terms of **a** and **b**.
Give your answer in its simplest form.

OB = a + b; BC = OC − OB = 2b − a

OD = (2/5)(a + b)

BE = (1/5)(2b − a)

OE = OB + BE = (4/5)a + (7/5)b

DE = OE − OD = (2/5)a + b

1 mark
D or E division ratio

1 mark
BC=2b−a; OD

1 mark
form OE−OD

1 mark
(2/5)a+b

(2/5)a + b

(Total for Question 18 is 4 marks)

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